

SECTION B

Answer **all** questions.

7. Read through the following article carefully.

Paragraph

A summary of binary stars

First of all, we call two stars orbiting their common centre of mass a “binary star”. Binary stars fall into 5 different classes, categorised by the detection method: 1

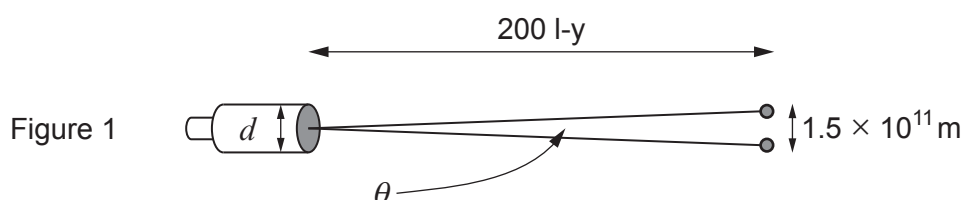
1. Visual binaries
2. Spectroscopic binaries
3. Eclipsing binaries
4. Astrometric binaries
5. Non-eclipsing binaries that can be detected using photometry.

Visual binaries

These do what it says on the tin – the angular separation of the stars is great enough that both stars can be seen as separate objects through a telescope. There is a simple relationship between the minimum angular separation θ (in radians), the diameter, d , of the telescope and the wavelength, λ , of the radiation used. 2

$$\theta = 1.22 \frac{\lambda}{d} \quad \text{Equation 1}$$

This equation means that a nearby binary star that is around 200 light years (l-y) away will only be visible as two stars with a $d = 8.0$ m telescope if the two stars are separated by at least the Earth-Sun distance of 1.5×10^{11} m (see Figure 1). 3



Perhaps the most famous of visual binary stars is the brightest star in the night sky – Sirius. Sirius is not, in fact, a single star but a binary consisting of Sirius A and Sirius B that orbit each other with a period of around 50 years. Sirius A is a main sequence star which has roughly 2 times the mass of the Sun, 1.7 times the Sun’s temperature and 25 times its luminosity. Sirius B is a white dwarf with a similar mass to our Sun, a surface temperature 4.3 times greater than our Sun but a luminosity 18 times less than our Sun. 4

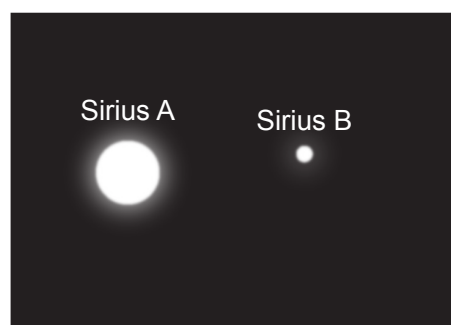


Figure 2



Spectroscopic binaries

These are binaries where the stars cannot be viewed separately but the existence of the two stars can be inferred from the variation of the Doppler shifted light from the system. Both stars will orbit their mutual centre of mass and hence their Doppler shifts will vary periodically during their orbits. This theory is the same as that used to discover exoplanets from the wobble of their parent star. 5

Eclipsing binaries

These happen when the line of sight of the observer is in the plane of the orbit of the two stars. When this happens, one star can pass directly in front of the other leading to a drop in the intensity of light entering the observer's telescope. An idealised graph of intensity against time is shown in Figure 3. 6

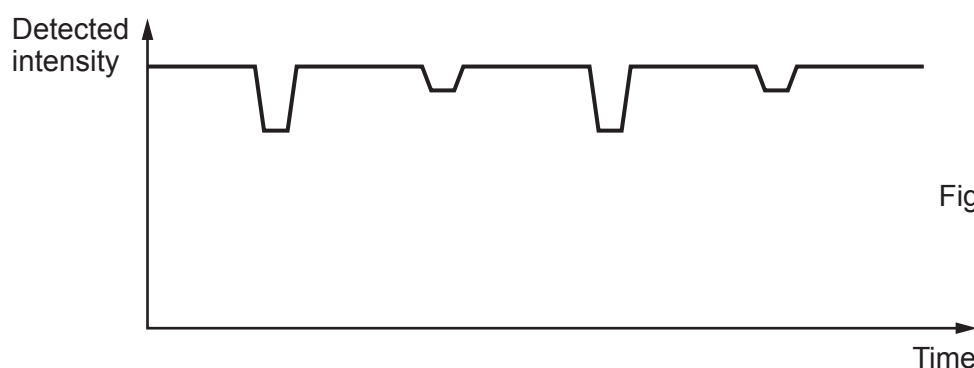


Figure 3

In general, there are two eclipses for each orbit. The larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star. 7

Astrometric binaries

These are binary systems where only one star is visible and that star can be seen moving in an orbit around something which is invisible. These are different from spectroscopic binaries because the visible star can actually be seen to move rather than inferring its motion from the variation of the Doppler shift. Stars orbiting neutron stars have been observed by this method but the ultimate goal of this technique would be to observe a star orbiting a black hole. 8

Non-eclipsing binaries that can be detected using photometry

This simply means detecting binaries by measuring light intensity but not via the eclipsing method described earlier. It turns out that there are three different techniques of finding binaries within this class of binary. First, an increased light intensity can be detected periodically from the binary system – this is caused by light being reflected from one star, off the other and into the Earth's telescope. Second, one star can exert such a strong gravitational pull on the other that it becomes deformed – this leads to a periodic fluctuation in the light intensity detected. Third, is an effect sometimes known as Doppler beaming – a star moving towards an observer will lead to an increased number of photons detected per second (as well as a decrease in the wavelength). 9

These 5 different classes of binary stars can provide an enormous amount of valuable data regarding stellar orbits and masses but they can also allow measurement of galactic distances and even give insights into black holes and supernovae. 10



Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

- (a) A light year is 9.46×10^{15} m. Use equation 1 and a wavelength of 500 nm to show that the author's claim about an 8 m telescope viewing the binary star mentioned in paragraph 3 and Figure 1 is correct. [4]

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- (b) (i) Show that the distance between Sirius A and Sirius B is approximately 3×10^{12} m (see paragraph 4). [Mass of the Sun is 2×10^{30} kg.] [2]

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- (ii) Show that the centre of mass of the Sirius binary system is located approximately 1×10^{12} m from Sirius A. [2]

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- (iii) Use data in paragraph 4 and Stefan's law to calculate the ratio of the radii of Sirius A and Sirius B:

[4]

$$\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}}$$

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- (c) (i) Evaluate whether the variation of light intensity shown in Figure 3 is consistent with an eclipsing binary system (see paragraph 6).

[3]

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- (ii) Use Stefan's law to explain why the "larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star" (see paragraphs 6 and 7 and Figure 3).

[2]

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- (d) Explain why a star orbiting a much more massive black hole is an astrometric binary (see paragraph 8). [1]

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- (e) One of the three methods for detecting non-eclipsing binaries that can be detected by photometry, suggests that stars do not behave as ideal black bodies. Explain which **one** of the three methods relies on stars not behaving as black bodies (see paragraph 9). [2]

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END OF PAPER

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